

## 1. Proof Insurance (Mandatory Class Proofs)

### Theorem 1: Alternate Variance Formula

*Proof.* By definition of variance and utilizing the linearity properties of mathematical expectation ( $\mathbb{E}$ ):

$$\begin{aligned}\text{Var}[X] &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2X \cdot \mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2X \cdot \mathbb{E}[X]] + \mathbb{E}[(\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X] \cdot \mathbb{E}[X] + (\mathbb{E}[X])^2 \quad (\text{Pull constants}) \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2.\end{aligned}\quad \square$$

### Theorem 2: Generalized Conditioning Rule (Chain Rule)

*Proof.* Expand the Right-Hand Side (RHS) using conditional probability definition  $P(A|B) = \frac{P(A \cap B)}{P(B)}$ :

$$\begin{aligned}\text{RHS} &= P(E_1) \cdot \frac{P(E_1 \cap E_2)}{P(E_1)} \cdot \frac{P(E_1 \cap E_2 \cap E_3)}{P(E_1 \cap E_2)} \cdots \\ &\quad \cdots \frac{P(E_1 \cap E_2 \cap \cdots \cap E_n)}{P(E_1 \cap E_2 \cap \cdots \cap E_{n-1})}\end{aligned}$$

Observe the telescoping cancellation where intermediate joint probability denominators cancel matching prior numerators:

$$\begin{aligned}\text{RHS} &= \cancel{P(E_1)} \cdot \frac{\cancel{P(E_1 \cap E_2)}}{\cancel{P(E_1)}} \cdots \frac{P(E_1 \cap \cdots \cap E_n)}{\cancel{P(E_1 \cap \cdots \cap E_{n-1})}} \\ &= P(E_1 \cap E_2 \cap \cdots \cap E_n) = \text{LHS}.\end{aligned}\quad \square$$

## Bonus Proof Asset Block

**Identity:**  $\mathbb{E}[cX] = c\mathbb{E}[X]$

*Proof.*  $\mathbb{E}[cX] = \sum (cx)f(x) = c \sum xf(x) = c\mathbb{E}[X].$   $\square$

**Identity:**  $\text{Var}[aX + b] = a^2 \text{Var}[X]$

*Proof.* Let  $\mu = \mathbb{E}[X]$ .  $\mathbb{E}[aX + b] = a\mu + b$ .  
 $\text{Var}[aX + b] = \mathbb{E}[((aX + b) - (a\mu + b))^2]$   
 $= \mathbb{E}[(aX - a\mu)^2] = \mathbb{E}[a^2(X - \mu)^2] = a^2 \text{Var}[X].$   $\square$

**Identity:**  $P(A^c) = 1 - P(A)$

*Proof.*  $A \cup A^c = S$  and  $A \cap A^c = \emptyset$ . By Axiom 3,  $P(A \cup A^c) = P(A) + P(A^c) \implies P(S) = P(A) + P(A^c)$ . By Axiom 2,  $1 = P(A) + P(A^c) \implies P(A^c) = 1 - P(A)$ .  $\square$

## Baseline Foundations & Set Axioms

### The 3 Core Axioms of Probability:

- **Axiom 1:**  $P(A) \geq 0$  (Probability cannot be negative)
- **Axiom 2:**  $P(S) = 1$  (The entire sample space is certain)
- **Axiom 3:** If  $A \cap B = \emptyset$ , then  $P(A \cup B) = P(A) + P(B)$

### Vital Derived Rules:

- **Complement Property:**  $P(A^c) = 1 - P(A)$
- **Inclusion-Exclusion Rule:**  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **DeMorgan's Laws:**  $(A \cup B)^c = A^c \cap B^c$  and  $(A \cap B)^c = A^c \cup B^c$
- **Null Probability:**  $P(\emptyset) = 0$

## Boundary & Inequality Translation Vocabulary

Let  $X$  be a discrete process approximated by a Normal curve with parameters  $\mu$  and  $\sigma$ .

- **"Fewer than  $k$ " / "Strictly less than  $k$ " ( $X < k$ ):**  
 Discrete Setup:  $P(X \leq k - 1)$   
 CLT Continuity Shift:  $P(X_{\text{normal}} \leq k - 0.5)$
- **"At most  $k$ " / "No more than  $k$ " ( $X \leq k$ ):**  
 Discrete Setup:  $P(X \leq k)$   
 CLT Continuity Shift:  $P(X_{\text{normal}} \leq k + 0.5)$
- **"More than  $k$ " / "Strictly greater than  $k$ " ( $X > k$ ):**  
 Discrete Setup:  $1 - P(X \leq k)$   
 CLT Continuity Shift:  $P(X_{\text{normal}} \geq k + 0.5)$
- **"At least  $k$ " / "No fewer than  $k$ " ( $X \geq k$ ):**  
 Discrete Setup:  $1 - P(X \leq k - 1)$   
 CLT Continuity Shift:  $P(X_{\text{normal}} \geq k - 0.5)$
- **"Between  $a$  and  $b$  inclusive" ( $a \leq X \leq b$ ):**  
 Discrete Setup:  $P(X \leq b) - P(X \leq a - 1)$   
 CLT Continuity Shift:  $P(a - 0.5 \leq X_{\text{normal}} \leq b + 0.5)$

**The Quick Rule of Thumb:** If the integer value  $k$  is *included* in your target set, your continuous boundary must swell outward by 0.5 to capture that entire bar. If  $k$  is *excluded*, your boundary must shrink away from it by 0.5.

## 2. Integral Calculus Moment Toolbox (Section 5.1)

When processing continuous random variables, polynomials scaled by exponential decay profiles require precise integration setups:

**Integration by Parts (IBP):**  $\int u dv = uv - \int v du$   
 Selection rule: Set Algebraic ( $x, x^2$ ) to  $u$ , Exponential ( $e^{-\lambda x}$ ) to  $dv$ .

- **Type A ( $\mathbb{E}[X]$  Mean Framework):**  $\int x e^{-\lambda x} dx$   
 Let  $u = x \implies du = dx$ ;  $dv = e^{-\lambda x} dx \implies v = -\frac{1}{\lambda} e^{-\lambda x}$ .

$$\int x e^{-\lambda x} dx = -\frac{x}{\lambda} e^{-\lambda x} - \frac{1}{\lambda^2} e^{-\lambda x}$$

- **Type B ( $\mathbb{E}[X^2]$  Second Moment Framework):**  $\int x^2 e^{-\lambda x} dx$   
 Let  $u = x^2 \implies du = 2x dx$ ;  $dv = e^{-\lambda x} dx \implies v = -\frac{1}{\lambda} e^{-\lambda x}$ .

$$\int x^2 e^{-\lambda x} dx = -\frac{x^2}{\lambda} e^{-\lambda x} + \frac{2}{\lambda} \int x e^{-\lambda x} dx$$

## 3. Master Scenario: Discrete PMF to CDF Staircase Graph

**The Setup:** Given a discrete PMF graph with support  $x \in \{0, 1, 3, 6\}$  and point mass probabilities  $f(0) = \frac{1}{5}$ ,  $f(1) = \frac{7}{20}$ ,  $f(3) = \frac{3}{10}$ , and  $f(6) = \frac{3}{20}$ . Construct the piecewise mathematical function and describe its graphical layout perfectly.

**Execution Workflow:** By definition,  $F(x) = P(X \leq x)$ , requiring a running cumulative total from left to right.

### 1. Accumulate Boundaries:

- For  $x < 0$ :  $F(x) = \mathbf{0}$
- For  $0 \leq x < 1$ :  $F(x) = f(0) = \mathbf{0.20}$
- For  $1 \leq x < 3$ :  $F(x) = 0.20 + f(1) = 0.20 + 0.35 = \mathbf{0.55}$
- For  $3 \leq x < 6$ :  $F(x) = 0.55 + f(3) = 0.55 + 0.30 = \mathbf{0.85}$
- For  $x \geq 6$ :  $F(x) = 0.85 + f(6) = 0.85 + 0.15 = \mathbf{1.00}$

### 2. Formal Piecewise CDF Notation Framework:

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 0.20 & \text{if } 0 \leq x < 1 \\ 0.55 & \text{if } 1 \leq x < 3 \\ 0.85 & \text{if } 3 \leq x < 6 \\ 1.00 & \text{if } x \geq 6 \end{cases}$$

### 3. Professor Gangaram's Strict Graphing Conventions:

- **Solid Dots (•):** Must be plotted at the absolute *left edge* of every single flat step segment (e.g., at (0, 0.20), (1, 0.55), (3, 0.85), and (6, 1.00)) to indicate inclusivity.

- **Open Circles (◦):** Must be plotted at the absolute *right edge* of every flat step segment (e.g., at (0, 0), (1, 0.20), (3, 0.55), and (6, 0.85)) to denote the non-inclusive interval cutoff boundary.
- **Asymptotes:** Leftmost line starts at  $-\infty$  resting flat on height 0. Rightmost line shoots out from (6, 1.00) heading straight to  $+\infty$  locked perfectly at height 1.00.

## 3. Master Scenario A: Polya Urn Sequence Transitions

**Problem Pattern:** Urn contains 2 Red, 3 Blue. Draw 3 times sequentially. Upon each selection, replace the drawn token *plus add 4 additional tokens of the opposite color*. Evaluate  $P(R_1 | B_3)$ .

**Execution Workflow:** Apply conditional definition:  $P(R_1 | B_3) = \frac{P(R_1 \cap B_3)}{P(B_3)}$ .

### 1. Track changing urn composition along paths starting with $R_1$ :

- Path  $R_1 R_2 B_3$ :  $\frac{2}{5} \rightarrow$  Add 4 Blue (Urn: 2R, 7B)  $\rightarrow \frac{2}{9} \rightarrow$  Add 4 Blue (Urn: 2R, 11B)  $\rightarrow \frac{11}{13}$ . Joint Path Value =  $\frac{44}{585}$ .
- Path  $R_1 B_2 B_3$ :  $\frac{2}{5} \rightarrow$  (Urn: 2R, 7B)  $\rightarrow \frac{7}{9} \rightarrow$  Add 4 Red (Urn: 6R, 7B)  $\rightarrow \frac{7}{13}$ . Joint Path Value =  $\frac{98}{585}$ .

### 2. Track changing urn composition along paths starting with $B_1$ :

- Path  $B_1 R_2 B_3$ :  $\frac{3}{5} \rightarrow$  Add 4 Red (Urn: 6R, 3B)  $\rightarrow \frac{6}{9} \rightarrow$  Add 4 Blue (Urn: 6R, 7B)  $\rightarrow \frac{7}{13}$ . Joint Path Value =  $\frac{126}{585}$ .
- Path  $B_1 B_2 B_3$ :  $\frac{3}{5} \rightarrow$  (Urn: 6R, 3B)  $\rightarrow \frac{3}{9} \rightarrow$  Add 4 Red (Urn: 10R, 3B)  $\rightarrow \frac{3}{13}$ . Joint Path Value =  $\frac{27}{585}$ .

### 3. Total Law of Probability (Denominator $P(B_3)$ ):

$$\frac{44+98+126+27}{585} = \frac{295}{585}.$$

### 4. Target Intersection Numerator ( $P(R_1 \cap B_3)$ ): $\frac{44+98}{585} = \frac{142}{585}$ .

### 5. Complete Division: $P(R_1 | B_3) = \frac{142/585}{295/585} = \frac{142}{295} \approx \mathbf{0.4814}$ .

#### 4. Continuous Variable Integrations (HW 10)

A variable is continuous if its CDF is continuous everywhere. Total area under the density curve must normalize to 1:  $\int_{-\infty}^{\infty} f(x)dx = 1$ .

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x)dx, \quad \mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 \cdot f(x)dx$$

#### 5. Master Scenario B: Piecewise Running-Total CDFs

**Problem Pattern:** Solve for the normalization constant  $c$  and construct the absolute piecewise CDF expression  $F(y)$  across all boundaries for:

$$f(y) = \begin{cases} cy^2 & \text{if } 0 \leq y < 2 \\ c(6-y) & \text{if } 2 \leq y \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

**Execution Workflow:**

1. Enforce standard area integration properties to isolate  $c$ :

$$\int_0^2 cy^2 dy + \int_2^6 c(6-y) dy = 1 \implies c \left[ \frac{y^3}{3} \right]_0^2 + c \left[ -\frac{(6-y)^2}{2} \right]_2^6 = 1$$

$$c \left( \frac{8}{3} - 0 \right) + c \left( 0 - \left( -\frac{16}{2} \right) \right) = 1 \implies c \left( \frac{32}{3} \right) = 1 \implies c = \frac{3}{32}$$

2. Construct  $F(y) = P(Y \leq y)$  by carrying forward accumulated area:

- For  $y < 0$ :  $F(y) = 0$ .
- For  $0 \leq y < 2$ :  $F(y) = \int_0^y \frac{3}{32}t^2 dt = \frac{1}{32}y^3$ .
- For  $2 \leq y \leq 6$ : Add the total historical area of the first piece ( $F(2) = \frac{1}{32}(8) = \frac{1}{4}$ ) directly to the current running integral:

$$\begin{aligned} F(y) &= \frac{1}{4} + \int_2^y \frac{3}{32}(6-t) dt = \frac{1}{4} + \frac{3}{32} \left[ 6t - \frac{t^2}{2} \right]_2^y \\ &= \frac{1}{4} + \frac{3}{32} \left( \left( 6y - \frac{y^2}{2} \right) - (12 - 2) \right) \\ &= -\frac{3}{64}y^2 + \frac{9}{16}y - \frac{11}{16} \end{aligned}$$

- For  $y > 6$ :  $F(y) = 1$ .

#### 6. Master Scenario C: Multi-Tier Special Distributions

**Problem Pattern:** Individual frames fail independently with  $p = 0.05$ . A cycle flags the stream if it catches *at least 2 errors across a frame block of 40 elements*. Find the probability that the 4th block is the first one flagged.

**Execution Workflow:**

1. Solve inner probability  $p^*$  using Binomial distribution ( $n = 40, p = 0.05$ ):

$$p^* = P(X \geq 2) = 1 - P(X \leq 1) = 1 - \text{binomcdf}(40, 0.05, 1) \approx 0.6010$$

2. Solve outer probability using Geometric distribution ( $k = 4, p^* \approx 0.6010$ ):

$$P(Y = 4) = (1 - p^*)^3 \cdot p^* = (0.3990)^3 \cdot 0.6010 = \mathbf{0.0382}$$

#### 7. Master Scenario D: CLT Continuity Adjustments

**Problem Pattern:** An application has an error rate of  $p = 0.20$ . Across 900 independent packet items, evaluate the probability that the final error count falls *between 170 and 195 inclusive*.

**Execution Workflow:**

1. Local parameters:  $\mu = np = 900(0.20) = 180$ .  
 $\sigma^2 = np(1-p) = 900(0.20)(0.80) = 144 \implies \sigma = 12$ .
2. Shift discrete integer boundaries outwards by  $\pm 0.5$  to account for histogram bar width under continuous normal models:

$$P(170 \leq X \leq 195) \approx P(169.5 \leq X_{\text{normal}} \leq 195.5)$$

3. Convert to standard Z-scores:

$$Z_{\text{low}} = \frac{169.5 - 180}{12} = -0.88; \quad Z_{\text{high}} = \frac{195.5 - 180}{12} = 1.29$$

4. Final area score:  $\text{normalcdf}(-0.88, 1.29, 0, 1) = \mathbf{0.7121}$ .

#### 8. Core Linear & Operational Shortcuts

- **Set Axioms (HW 1–2):** DeMorgan's:  $(A \cup B)^c = A^c \cap B^c$ .  
Inclusion-Exclusion:  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ .
- **Linear Operations (Section 4.2–4.3):** Expectation is linear:  $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ . Variance scales by the square and drops translation constants entirely:  $\text{Var}[aX + b] = a^2\text{Var}[X]$ .
- **Staircase CDF (Section 4.1):** Discrete step height jump sizes indicate localized point probability values:  $P(X = x_i) = F(x_i) - F(x_{i-})$ .